

User Study Results

CLONE UI Panel - Week 9 Results Summary

Participants 8 users tested the CLONE plugin and completed the feedback form.	Overall rating 62.5% Good, 37.5% Neutral, 0% negative.	Most useful feature Diff Preview was selected most often.
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1. Evaluation Goal

The user study was conducted to evaluate whether the CLONE interface helps developers understand a detected clone, review the proposed Extract Method refactoring, interpret validation evidence, and decide whether to apply, edit, or reject the suggestion. The study focused on usability, reviewability, trust support, and developer control.

2. Data Overview

Measure	Result	Interpretation
Number of participants	8	Small exploratory user study for final project evaluation.
Overall experience	62.5% Good, 37.5% Neutral	All responses were neutral or positive; no negative ratings were reported.
Most useful feature	Diff Preview (3 users)	Users most valued seeing the proposed code change before applying it.
Second most useful feature	Details and Provenance (2 users)	Users also valued audit-style information about the suggestion.
Other selected features	Explanation section, Multiple-clone dialog, Action buttons (1 each)	Usefulness was distributed across several UI components.

3. Overall Experience

The overall response was positive. Five out of eight users rated the CLONE UI Panel as "Good," while three users rated it as "Neutral." No user selected a negative rating. This suggests that participants generally found the interface acceptable and usable, while the neutral ratings indicate that additional refinement is still needed.

How would you rate your overall experience with the CLONE UI Panel?

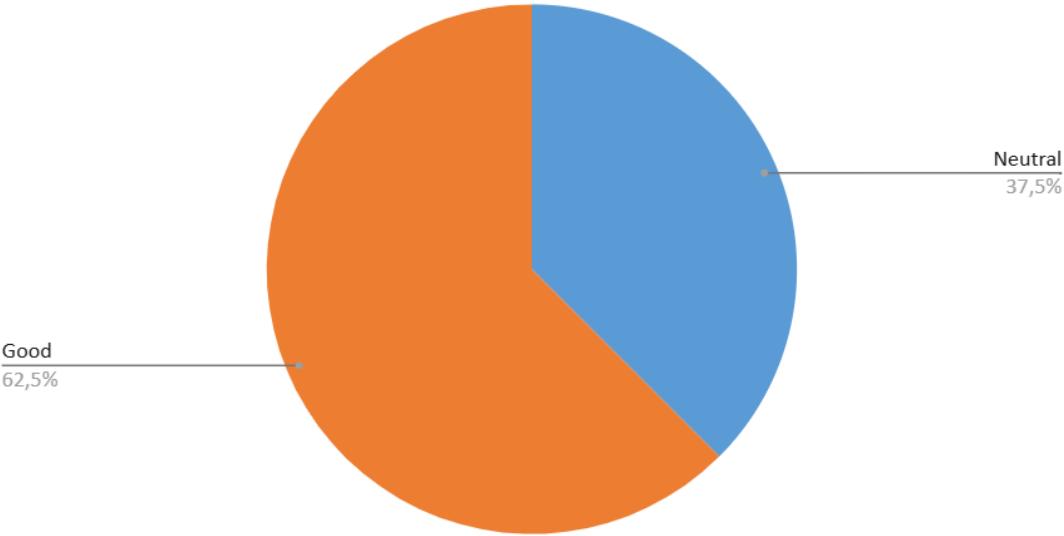


Figure 1. Overall user experience rating for the CLONE UI Panel.

4. Most Useful Feature

The Diff Preview was the most frequently selected useful feature, chosen by three users. Details and Provenance was selected by two users. The Explanation section, Multiple-clone dialog, and Action buttons were each selected by one user. This indicates that users especially valued features that made the refactoring visible, reviewable, and easier to inspect before applying source-code changes.

Which feature was the most useful?

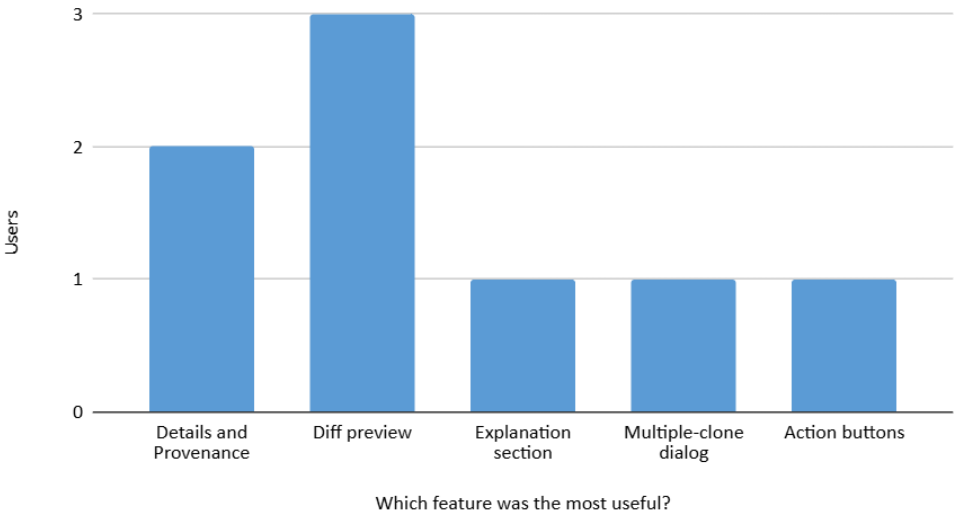


Figure 2. User responses for the most useful CLONE UI feature.

5. Key Findings

- **Reviewability was the strongest theme.** Users most often selected the Diff Preview, showing that the ability to compare original duplicated code with the proposed refactoring was central to understanding the suggestion.
- **Transparency also mattered.** The Details and Provenance feature was the second most selected, suggesting that users valued additional evidence about where the suggestion came from and what code was involved.
- **Developer control remained important.** The multiple-clone dialog and action buttons were selected by some users, showing that inclusion/exclusion and apply/edit/reject controls still contribute to the review process.
- **The experience was acceptable but not finished.** The absence of negative ratings is promising, but the neutral ratings show that setup, validation clarity, and error handling can still be improved.

6. Limitations and Future Work

The user study also revealed limitations. Some users experienced technical issues such as API key problems, model/provider errors, setup difficulty, or cases where the workflow did not fully complete. In addition, some parts of the interface, such as AI Trust and validation information, the multiple-clone dialog, or action buttons, may need clearer labels and stronger visual guidance. Future work should focus on simplifying setup, clarifying validation states, improving failure messages, and making the interface easier to understand for first-time users.